

INTERNSHIP PROJECT REPORT

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Project Name: ESP32 Ultrasonic Distance & Intrusion Detection Node

Domain: Internet of Things (IoT) Internship

1. PROJECT OVERVIEW

1.1 System Architecture & Distance Monitoring

This project implements an industrial-grade Autonomous Security & Distance Monitoring Node centered on the ESP32 microcontroller. The system triggers an HC-SR04 ultrasonic sensor, measures return echo travel time via pulse-width modulation, and computes real-time distance telemetry to detect objects crossing a predefined security boundary.

1.2 Engineering Highlights & Innovations

- Voltage Divider Level Shifting: Employs a precision resistor network (1 k Ω and 2.2 k Ω) to safely step down the 5V HC-SR04 Echo output signal to a secure 3.3V logic level compatible with ESP32 input pins.
- Non-Blocking Timing Architecture: Utilizes millis()-based interval scheduling to handle continuous sensor polling without freezing core execution threads.
- Real-Time Boundary Alerting: Applies strict threshold logic (< 50 cm) to instantly flag proximity breaches and output diagnostic telemetry to the Serial Monitor.
- Optimized GPIO Interfacing: Integrates dedicated digital pins (GPIO 5 for Trigger and GPIO 18 for Echo) for accurate signal transmission and reception.

2. CIRCUIT PIN CONNECTIONS

HC-SR04 Ultrasonic Sensor:

- VCC -> ESP32 5V (Power Supply)
- GND -> ESP32 GND (Ground)
- TRIG -> ESP32 GPIO 5 (Digital Trigger Output)
- ECHO -> Voltage Divider (1 k Ω & 2.2 k Ω) -> ESP32 GPIO 18 (Digital Echo Input)

3. SIMULATION LINK & PROOF

Wokwi Live Simulation Link:

<https://wokwi.com/projects/471852870758591489>

Simulation Screenshot:

WOKWI

SAVE

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ESP32-Ultrasonic-Security-Node

Docs

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sketch.ino

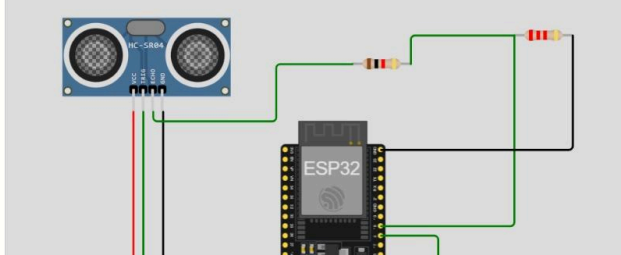
diagram.json

Library Manager

```
1 #include <Arduino.h>
2
3 // Pin Definitions
4 #define TRIG_PIN 5
5 #define ECHO_PIN 18
6
7 // Speed of sound in cm/us
8 #define SOUND_SPEED 0.0343
9
10 long duration;
11 float distanceCm;
12
13 unsigned long lastTime = 0;
14 const unsigned long timerDelay = 1000;
15
16 void setup() {
17   Serial.begin(115200);
18   pinMode(TRIG_PIN, OUTPUT);
19   pinMode(ECHO_PIN, INPUT);
20
21   Serial.println("=====");
22   Serial.println(" Cloud-Connected Security Node Started ");
23   Serial.println("=====");
24 }
25
26
27 void loop() {
28   if (millis() - lastTime >= timerDelay) {
29     lastTime = millis();
30
31     // Trigger pulse creation
32     digitalWrite(TRIG_PIN, LOW);
33   }
34 }
```

Simulation

00:06.001 73%



Cloud-Connected Security Node Started
=====

Sensor Telemetry -> Distance: 403.45 cm
Sensor Telemetry -> Distance: 403.49 cm
Sensor Telemetry -> Distance: 403.45 cm
Sensor Telemetry -> Distance: 403.45 cm
Sensor Telemetry -> Distance: 403.45 cm

Activate Windows
Go to Settings to activate Windows.